

Ground return planes should be solid.

Shielding. Employ shielding techniques, such as ground planes and guard traces, to isolate sensitive analogue signals from noisy digital environments.

Power conversion and motor controller circuits

Power conversion circuits, such as direct current (DC)-DC converters and motor controllers, are notorious sources of EMI due to their switching nature.

Component placement. Place switching components (transistors, relays) close to each other to minimise the loop area and reduce radiated EMI.

Filtering. Use capacitors and ferrite beads to filter high-frequency noise at power inputs and outputs.

Shielding. Enclose power conversion circuits in metal shields to contain radiated EMI.

Sensitive RF receiver circuits

Sensitive radio frequency (RF) receiver circuits, such as those used in global positioning systems (GPS), cellular, or Wi-Fi devices, are particularly vulnerable to EMI.

RF isolation. Use dedicated RF ground planes to isolate RF sections from other noisy parts of the PCB.

Component placement. Place RF components away from high-speed digital circuits and switching power supplies.

Decoupling and filtering. High-quality decoupling capacitors and RF chokes are used to filter power supplies and signal lines feeding RF components. Noise occurs through parasitic coupling between noise sources and circuits or between circuits and free space.

Ground return planes

Ground return planes are essential for controlling EMI. They provide

a low-impedance path for return currents and help shield against EMI.

Continuous planes. Ensure ground planes are continuous and unbroken, avoiding slots and gaps that can increase EMI.

Ground vias. Multiple vias connect ground planes on different layers to maintain low impedance and improve EMI performance.

Layer stacking. Place ground planes adjacent to signal layers to create controlled-impedance transmission lines and reduce crosstalk and EMI.

3D electromagnetic simulation

Three-dimensional (3D) electromagnetic (EM) simulation tools are invaluable for predicting and reducing EMI in PCB designs. These tools allow designers to model the electromagnetic fields generated by PCB traces and components, identify potential EMI sources, and optimise the board layout accordingly.

Simulation software. Use simulation software, such as Ansys HFSS or CST Microwave Studio, to analyse the electromagnetic behaviour of the PCB.

Field analysis. Perform field analysis to visualise how electric and magnetic fields propagate and interact within the PCB.

Optimisation. Iterate the design based on simulation results to minimise EMI and improve signal integrity.

Some key tips for PCB designers to minimise EMI

Ensure a continuous, unbroken ground plane. Avoid splits or slots that can disrupt current return paths and increase EMI. Placing power and ground planes next to each other forms a high-frequency decoupling capacitor, which helps suppress EMI.

Use microstrip (signal over a

ground plane) or stripline (signal between two ground planes) techniques for routing high-speed signals to control electromagnetic fields.

For high-speed or sensitive signals, use differential pairs. These can cancel out common-mode noise and reduce EMI. Ensure differential pairs are matched in length and maintain consistent spacing to preserve the signal's integrity and minimise EMI.

Use metal shields or cans over particularly noisy or sensitive components to contain or block EMI.

Grounding techniques

Star grounding. Use a star grounding scheme where all grounds connect to a single point to prevent ground loops.

Chassis grounding. Connect the PCB ground to the chassis ground at a single point to avoid ground loops and improve EMI performance.

Use surface mount devices (SMDs) instead of leaded devices. This is particularly critical for two-layer or four-layer circuit boards. Place components that interact with each other closely together to minimise trace lengths.

Physically separate noisy components, such as oscillators or switching regulators, from sensitive analogue or RF circuits. Conduct thorough EMI testing, such as electromagnetic compatibility (EMC) compliance tests, on prototypes to identify and address any issues before mass production.

Designing PCBs to manage EMI requires a comprehensive approach that carefully plans various factors. 3D EM simulation is critical in identifying and mitigating potential EMI sources. By implementing these strategies, designers can create robust PCBs against EMI, ensuring reliable performance in various applications, from power conversion circuits to sensitive RF receivers. **EFY**